

AMC 10 Crash Course • Test T4 • Counting and Probability

10 problems • 30 minutes • no calculator • mastery bar 7 of 10

Name: _____

Date: _____

1. How many even four-digit numbers can be formed from the digits 1, 2, 3, 4, 5 with no digit repeated?
(A) 24 (B) 48 (C) 60 (D) 96 (E) 120
2. Six people line up in a row. Ann and Ben must stand next to each other. How many arrangements are there?
(A) 120 (B) 144 (C) 240 (D) 480 (E) 720
3. A team of 3 boys and 2 girls is chosen from 5 boys and 4 girls. In how many ways?
(A) 20 (B) 40 (C) 54 (D) 60 (E) 90
4. How many diagonals does a convex 15-gon have?
(A) 90 (B) 100 (C) 105 (D) 120 (E) 135
5. How many integers from 1 to 200 are divisible by 6 or by 8?
(A) 40 (B) 42 (C) 46 (D) 48 (E) 50
6. How many four-digit numbers contain at least one digit 5?
(A) 2916 (B) 3096 (C) 3168 (D) 3268 (E) 5832
7. Two dice are rolled. What is the probability that the sum is at least 10?
(A) $\frac{5}{36}$ (B) $\frac{1}{6}$ (C) $\frac{7}{36}$ (D) $\frac{2}{9}$ (E) $\frac{1}{4}$
8. Two cards are drawn without replacement from a standard 52-card deck. What is the probability that they are of the same suit?
(A) $\frac{4}{17}$ (B) $\frac{1}{4}$ (C) $\frac{5}{17}$ (D) $\frac{6}{17}$ (E) $\frac{1}{3}$
9. How many paths lead from (0, 0) to (5, 4) by unit steps right and up?
(A) 84 (B) 100 (C) 120 (D) 126 (E) 210
10. A coin is tossed 5 times. What is the probability that no two heads come up in a row?
(A) $\frac{3}{8}$ (B) $\frac{13}{32}$ (C) $\frac{7}{16}$ (D) $\frac{15}{32}$ (E) $\frac{1}{2}$

Answer Key and Review

Parent page: cut off or do not print for the student. Map misses to lessons: 1-2 → 4.1, 3-4 → 4.2, 5-6 → 4.3, 7-8 → 4.4, 9-10 → 4.5.

1	2	3	4	5	6	7	8	9	10
B	C	D	A	E	C	B	A	D	B

1. Last digit 2 or 4: $2 \cdot 4 \cdot 3 \cdot 2 = 48$.
2. Glue: $2 \cdot 5! = 240$.
3. $C(5, 3) \cdot C(4, 2) = 10 \cdot 6 = 60$.
4. $C(15, 2) - 15 = 105 - 15 = 90$.
5. $33 + 25 - 8 = 50$ (the intersection is multiples of the lcm, 24).
6. Complement: 9000 four-digit numbers in total, $8 \cdot 9^3 = 5832$ with no 5; answer $9000 - 5832 = 3168$.
7. Sums 10, 11, 12 give $3 + 2 + 1 = 6$ pairs out of 36: $1/6$.
8. The second card from the first card's suit: $12/51 = 4/17$.
9. $C(9, 4) = 126$.
10. No two heads in a row: 13 sequences out of 32 (Fibonacci: 2, 3, 5, 8, 13): $13/32$.